



Faculty of Business and Law

Assessment Task – Assignment 2

INF30035

Business Process Analysis

“Analysing and Designing To-be Process Modelling”

Due: Friday, 18th Oct. 2024 @ 23:59 AEST

Assignment 2 Weight = 40%

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Executive Summary

This report provides a comprehensive analysis of Business Process Management (BPM) and its application to improve operations at SGC. It begins by defining BPM and exploring current trends, emphasizing its importance in enhancing organizational performance. The analysis then focuses on SGC's specific challenges, identifying key bottlenecks such as process delays, communication inefficiencies, resource wastage, and role overlaps. To address these problems, the report proposes several best practices including contact reduction, split responsibilities, re-sequencing, knockout, and resource centralization. These practices are then applied to SGC's specific operational issues, detailing how they can resolve current inefficiencies. The report also outlines a change management plan, including staff training, phased implementation, and ongoing monitoring to ensure the successful adoption of the new processes. Level 2 Business Process Modeling is used to illustrate both the current (AS-IS) and proposed (TO-BE) operational models. Finally, an automation model using Business Process Management Systems (BPMS) is presented to further streamline SGC's operations.

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Part 1: Business Process Management and Improvement

In the past, information technology primarily supported individual tasks and data storage. Today, business processes and information systems are deeply interconnected, with processes relying heavily on IT systems, which in turn are shaped by the processes they support (van der Aalst, 2009). Business Process Management (BPM) focuses on this interplay. The Australian BPM Community of Practice defines BPM as a “structured, coherent and consistent way of understanding, documenting, modelling, analyzing, simulating, executing, measuring and continuously changing end-to-end business processes and all involved resources in light of their contribution to business improvement” (ABPMP, n.d). Key BPM knowledge areas encompass process analysis, design, and automation for managing operations and organizing workflows (van der Aalst et al., 2016).

A study on BPM scientific output trends revealed positive developments in the Management network. Key concepts include BPM management, process and organizational performance, and operations management. Notably, there is a significant uptrend in research on Business Process Models and Notation, highlighting its growing importance in academic process modelling (Reijers, 2021). In the industry, recent BPM trends focus on enhancing intelligence and collaboration. Intelligent BPM (iBPM) integrates real-time reporting, allowing flexible, goal-driven process models with immediate adjustments during execution. This represents a shift from traditional post-execution reporting, enabling dynamic process management (von Rosing, 2015). Social BPM, a collaborative approach to process design, integrates social technologies and emphasizes the perspectives of both process doers and receivers, fostering continuous learning and improvement. It involves process integration between customers and employers through platforms like Facebook or “X” (Gartner, 2014). Additionally, there is a growing demand for BPM certification to ensure experienced personnel have comprehensive, vendor-neutral skills. These certifications cover business process principles, modelling techniques, and architectural layer modelling, equipping personnel to lead BPM initiatives effectively (von Rosing, 2015).

BPM is a critical driver of business performance. By optimizing end-to-end processes across functions and beyond enterprise boundaries, companies can achieve operational excellence and adapt to market changes. BPM methodologies make processes visible through modelling,

monitoring, and simulation, allowing both business and IT stakeholders to understand and improve them (von Rosing, 2015). BPM also offers significant advantages, including process automation, flexible IT system integration, and enhanced transparency. These benefits improve operational efficiency, system interoperability, and process visibility, enabling businesses to streamline operations and adapt more effectively to changing requirements (Dumas et al., 2018). BPM synchronizes process models with execution, allowing businesses to optimize performance and gain a competitive edge in today's dynamic environment (von Rosing, 2015).

Part 2: Business Process Improvement Recommendation

1) Resource and Performance

Process Bottlenecks

A bottleneck in a company represents a critical point where tasks accumulate, impeding the normal flow of production and potentially disrupting operations. These bottlenecks emerge when the amount of tasks exceeds the organization's capacity to respond effectively. Left unaddressed, bottlenecks negatively impact competitiveness and customer satisfaction (Cardarello, 2024). For SGC, bottlenecks significantly reduce productivity, even if other processes function efficiently. These constraints create delays and idle time, diminishing overall workflow efficiency.

At SGC, the main bottleneck that arises during coach allocation lies in matching these requirements with the coaches' skills, expertise, and availability. Coordinating various tasks, schedules, and coaching strengths complicates this already time-consuming process. Moreover, repeated verification loops between clients and coaches exacerbate delays, leading to multiple iterations caused by scheduling conflicts or changes in client preferences. These delays extend session delivery times and increase friction, resulting in customer dissatisfaction and potentially damaging SGC's reputation.

Addressing this bottleneck is crucial for streamlining operations, enhancing customer experience, and maintaining a competitive edge in the coaching industry. The impact of bottlenecks includes lost opportunities, reduced client engagement, and resource strain as staff struggle to manage backlogs. Furthermore, uncertainty in scheduling may discourage potential clients from committing to SGC's services, hindering growth and market penetration.

Resource Utilisation and Performance Improvements

To achieve 80% utilization and better organise staff allocation, SGC needs a multi-faceted approach. Implementing a robust scheduling system to track coach availability, skills, and client preferences in real-time is essential. This technology-driven solution, combined with flexible working hours and data analytics to predict peak periods, can optimize resource allocation (Vulpen, 2021). Cross-training coaches in multiple areas of expertise will also increase versatility and streamline allocation (Bleich, 2018). Regular performance reviews and utilization analyses will identify areas for improvement while employing part-time or on-call coaches during peak times can maintain high utilization rates. Studies show 70% of businesses experience increased productivity with temporary staff, and temporary staffing can reduce labour costs by 20% (Pivotal Solutions, 2024). Combining these strategies will help SGC effectively balance resource use with high-quality client service.

2) Issues Identified

Through an in-depth operational analysis of SGC, several key issues have emerged, impacting the efficiency of processes and the overall quality of service delivery. These challenges stem from inefficiencies in communication, workflow management, resource allocation, and data handling. The primary issues identified are as follows:

1. *Process Delays Due to Manual Interventions*

SGC's heavy reliance on manual processes, such as client inquiries, coach assignments, and payment processing, leads to delays and inefficiencies. To address this, implementing automation tools can significantly reduce manual verification and form processing. For example, automating coach scheduling and client onboarding will eliminate bottlenecks, speed up response times, and allow staff to focus on high-value tasks. This aligns with Lean's focus on reducing waste and Six Sigma's emphasis on improving process accuracy and efficiency (Lucas, 2019).

2. *Communication and Workflow Inefficiencies*

Task misalignment and fragmented communication in processes such as coach assignment and payment processing result in considerable delays. The absence of automation leads to delayed coach payments and client invoicing, resulting in bottlenecks. Communication

deficiencies among staff, clients, and coaches result in delayed responses, follow-ups, and miscommunication, thereby disturbing service timelines and adversely affecting client satisfaction (Specification Management, 2023).

3. *Resource Wastage and Ineffective Management*

The mismanagement of human resources and coach availability is a significant issue at SGC. Coaches are frequently underutilised due to delays in matching clients, while staff time is wasted on repetitive tasks such as manually processing forms, rechecking information and verifying. This poor creates scheduling gaps, reducing the number of sessions conducted. This inefficient use of resources increases operational costs and affects service quality (Chron Contributor, 2020), as coaches are not promptly matched with clients, leading to missed opportunities for both parties.

4. *Duplication of Efforts and Role Overlap*

Without a centralised system to manage client and coach information, staff duplicate efforts by manually re-entering data, increasing errors and delays. The duplication of effort wastes valuable time and undermines the reliability of SGC's data, ultimately affecting decision-making and the overall client experience (Dagher, 2021). Moreover, overlapping roles among staff members create confusion, diverting resources from improving services and undermining SGC's operational effectiveness (Shin & Konrad, 2017).

3) Selected Best Practices

A collection of best practices aligned with SGC's strategic goals can alleviate operational inefficiencies. Informed by a comprehensive analysis using the Devil's Quadrangle framework (see Appendix A), these practices focus on enhancing service quality while ensuring flexibility in the highly competitive and dynamic coaching industry. SGC must prioritise its ability to adapt to evolving client needs and industry trends to remain competitive and maintain operational efficiency.

The following best practices target SGC's primary challenges and aim to improve overall efficiency by aligning process enhancements with the company's strategic goals:

A. Contact Reduction

Minimising the number of interactions or handoffs during processes is essential to avoid delays and miscommunication. At SGC, excessive staff involvement in client follow-ups and coach coordination creates bottlenecks. Streamlining communication channels will improve efficiency and service delivery (Dumas et al., 2013).

B. Split responsibilities

Establishing distinct roles and responsibilities can optimise resource utilisation and prevent task duplication. Research by Bourne et al. (2017) shows that clearly defined job roles enhance productivity and reduce operational delays. By assigning specific functions to individuals, SGC can ensure that employees focus on their areas of expertise, leading to a more efficient workflow.

C. Re-sequencing

Re-sequencing involves restructuring the order of tasks to improve workflow efficiency and eliminate bottlenecks. By rearranging specific tasks within SGC processes, delays can be minimised, resulting in smoother transitions and enhanced experiences for clients and coaches (Indeed Team, 2024).

D. Knockout

This practice focuses on the early identification and elimination of tasks unlikely to succeed, preventing wasted time and resources (Dumas et al., 2013). For SGC, this could involve identifying potential issues, such as unpaid invoices or unresponsive coaches, early in the process to streamline operations and optimize resource allocation.

E. Resource Centralisation

Resource Centralization focuses on consolidating information and resources into a single platform to enhance accessibility and efficiency (Dumas et al., 2013). This ensures that all SGC's team members have access to up-to-date information, improving communication and reducing operational inefficiencies.

4) Best Practices Application

The following section explains how these best practices can be applied to solve the primary issues identified in the SGC "To-Be" model. A summarized table can be found in Appendix C.

1. Process Delays Due to Manual Intervention

A. Contact Reduction

- **Application in the To-Be Model:**

The updated process model will introduce a centralised inquiry form on the SGC website to collect all required client information upfront. This will include personal details, coaching requirements, referral codes, communication preferences (e.g., SMS or email), and client consent. Automating this inquiry process eliminates the need for receptionists to manually ask for or follow up on client information, reducing delays in onboarding and streamlining the initiation of the coaching assignment.

- **Outcome:**

Centralising enquiries reduces manual intervention and speeds up client onboarding (The Holaspirt Team, 2023). Client satisfaction will increase with streamlined client-staff communication, reducing follow-ups and miscommunication. As onboarding goes more easily, client satisfaction rises. Since they can use the existing layout, existing clients can save time re-entering personal information. For Emily's clients, the referral code will facilitate easy tracking of her clients. However, ongoing updates to the website and maintenance fees may pose a challenge. Additionally, clients might contact receptionists for help with completing the form, necessitating occasional manual assistance. Despite these challenges, the overall outcome is improved efficiency and minimised delays.

E. Resource Centralisation

- **Application in the To-Be Model:**

Resource centralisation entails the extensive enhancement of SGC's database to incorporate all pertinent coach details into a single, cohesive system. This centralized database will contain vital information, including coach profiles, specialties, availability schedules, areas of expertise, and additional relevant data. By consolidating this information, coach specialists will gain real-time access to essential details, eliminating the need for manual searches across

various platforms or documents. The centralized database will be designed for user-friendliness, allowing staff to quickly retrieve and update coach information as required. Moreover, the database will be continually updated, enabling the seamless addition of newly hired coaches, and ensuring that the information remains current and comprehensive.

- **Outcome:**

The advantages of implementing a centralised database include enhanced efficiency, as staff will no longer need to manually update or verify information. This centralisation provides accurate, real-time access to client and coach details, improving decision-making processes and minimising errors. However, establishing this requires a significant upfront investment in technology and training (IFP, 2023). Furthermore, the transition period may cause temporary disruptions to operations, potentially impacting service delivery during the implementation phase.

2. *Communication and Workflow Inefficient*

A. Contact Reduction

- **Application in the To-Be Model:**

Automated reminders and notifications will be implemented to send timely updates and confirmations to clients and coaches. Automation reduces the need for manual back-and-forth communication, preventing important information from being delayed or lost. Consolidating communication through a self-service portal ensures that clients are contacted only when necessary, minimising redundant messages.

- **Outcomes:**

Streamlining communication speeds up responses, clarifies information, and reduces delays. As a result, clients have a more effective communication process that lowers interactions and improves quality, making their experience smoother. However, a reduced direct connection may make some clients feel detached, forcing SGC to balance operational efficiency with personalised client care to retain good client relationships.

C. Re-sequencing

- **Application in the To-Be Model:**

Key tasks will be re-sequenced to improve process efficiency. For example, before a coaching assignment is confirmed, clients will need to complete a payment deposit. SGC will send clients an email after assigning a coach, outlining coaching session details and payment terms. Clients will be required to pay a 20% deposit before the session begins, with the remaining 80% to be paid upon session completion. This also serves as confirmation that the client agrees to the policies and has committed to the session.

- **Outcomes:**

The upfront deposit simplifies payment processing and reduces the risk of clients not paying after sessions. Taking a 20% payment beforehand reduces bad debt and encourages client commitment to sessions. SGC's debt collection procedures can also benefit from clients' email policy confirmations. However, the deposit requirement may inconvenience certain clients, affecting their willingness to use SGC services. Balancing this requirement with client satisfaction is crucial.

3. *Resource Wastage and Ineffective Management*

D. Knockout

- **Application in the To-Be Model:**

Proactive case filtering will be used to identify cases unlikely to proceed, such as clients who fail to provide necessary documentation or coaches who do not respond within a set timeframe. Additionally, the system will automatically flag clients on a blacklist and reject assignments accordingly.

- **Outcomes:**

Knockout improves resource allocation by allowing staff to focus on viable cases. By filtering out non-viable clients early in the process, staff can concentrate their efforts on high-priority cases that are more likely to succeed. However, rejected clients may feel dismissed, leading to dissatisfaction. Clear communication about the knockout criteria is essential to preserve client trust.

E. Resource Centralisation

- **Application in the To-Be Model:**

The To-Be model incorporates a unified database for both client and coach information, effectively centralising all relevant data into a single system. This comprehensive database will allow personnel to access real-time, correct data, reducing complicated data entry and verification. This centralisation also optimises coaching session scheduling by giving immediate coach availability, allowing staff to match clients with suitable coaches based on their needs. Centralised coach availability data reduces scheduling gaps and eliminates resource underutilization. By automating this procedure, SGC can improve operational efficiency and quickly match clients with trainers.

- **Outcome:**

Staff will no longer need to manually update or check information, improving efficiency. This centralisation reduces errors and enhances decision-making. However, the high upfront cost and any operational disruptions during the transition must be managed to avoid service delivery issues.

4. Duplication of Efforts and Role Overlap

B. Slit Responsibilities

- **Application in the To-Be Model:**

Clear roles and responsibilities will be defined within the SGC team to eliminate task duplication and overlap. Each team member will be assigned specific duties ensuring tasks are completed efficiently without redundancy (refer to Appendix B for job descriptions).

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- **Outcomes:**

Clearly defined roles lead to operational efficiency and collaboration by avoiding confusion and overlap. This also enables staff to focus on higher-priority tasks. However, some personnel may oppose changes in their duties and need training and time to adjust. Effective change management is important for a smooth transition.

5) Methodologies

Lean Six Sigma is an effective methodology for addressing SGC's process challenges, merging Lean's focus on efficiency with Six Sigma's emphasis on quality improvement. This

approach is vital for reducing inefficiencies in coach allocation, resource management, and manual interventions.

Lean Methodology

Lean focuses on eliminating waste by removing non-value-added activities (Terry, 2023). For SGC, this applies to reducing delays and manual tasks in coach assignments and client onboarding. Automating scheduling and using a centralised platform can streamline operations, saving time and resources. Lean's customer-centric approach aligns with SGC's goal to improve client satisfaction by enhancing process efficiency.

Six Sigma Methodology

Six Sigma aims to reduce process variation and defects through data-driven decision-making (Evans & Lindsay, 2015). For SGC, it can be used to address communication breakdowns, workflow misalignments, and task duplication. Applying tools like DMAIC (Define, Measure, Analyse, Improve, Control) can help SGC analyse inefficiencies, implement standardizations, and maintain consistent service quality.

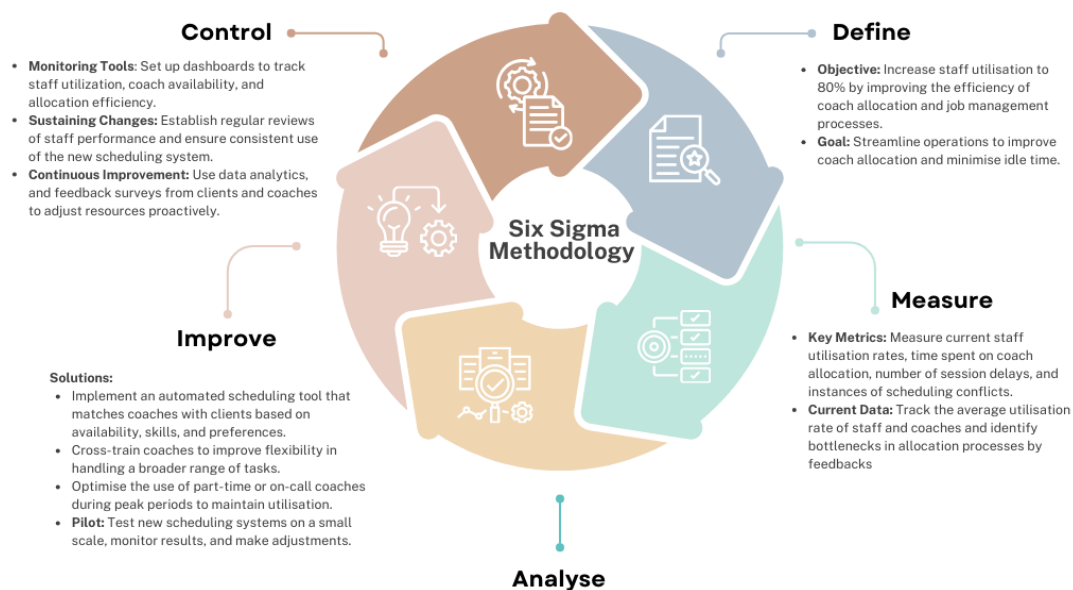


Figure 1. The DMAIC Cycle (Six Sigma Methodology) of SGC

Lean Six Sigma Integration

The combination of Lean and Six Sigma maximizes efficiency and quality (What is Lean Six Sigma?, 2013), ideal for SGC's operational needs. Lean Six Sigma can streamline coach allocation and payment processes while also reducing variability through data analysis. This ensures both faster service delivery and improved client satisfaction.

6) Limitations

This project has several limitations regarding what cannot be fully addressed or achieved:

- **Lack of Automation Technology Discussion:** The project does not thoroughly explore specific automation technologies that could enhance operational efficiency, making an in-depth analysis of available options and their implementation necessary.
- **Insufficient BPMN Detail:** The BPMN representation fails to capture the complexities of the form-filling process, limiting clarity on data collection and processing during client onboarding.
- **Unclear Coaching Pool Expansion Process:** Strategies for expanding the coaching pool are inadequately illustrated, which may hinder the understanding of effective methods for increasing available coaches.
- **Overlap in Best Practices:** Many proposed solutions share common elements, leading to ambiguity in understanding how each best practice uniquely addresses specific challenges.

7) Change of Management Plan

The proposed changes for SGC will follow the tuning approach, which focuses on making incremental adjustments to improve performance without requiring significant structural changes (Chaffey & White, 2011). Rather than overhauling existing processes, the plan involves adding a self-service portal, a centralized inquiry form, and automated communication tools to enhance current operations. This approach is designed to optimize processes by "doing things better," not radically changing the business model.

Implementing these gradual changes will increase operational efficiency, reduce delays, and improve client satisfaction without causing major disruptions (Amanatullah, 2023). By avoiding drastic shifts, SGC can ensure smooth transitions, minimizing resistance from staff and clients. The tuning approach aligns with SGC's goals of maintaining competitiveness, enhancing service quality, and ensuring flexibility while preparing for future growth.

A well-structured change management plan will be essential, focusing on staff training, phased implementation, and continuous monitoring to align with the TO-BE model and support future growth.

Staff Training

Training staff is vital to prepare employees for managing the new systems and processes. The training will focus on optimizing resource allocation and improving staff performance, targeting an 80% utilisation rate. Key steps include:

- **Training Needs Assessment** to identify skill gaps.
- **Customised Training Programs** focusing on technical, process, and customer support skills.
- **On-the-job Support** to provide ongoing assistance after the training.
- **Evaluation and Feedback** to refine the training and build staff confidence in using the new systems.

Implementation Phases

The transition to the To-Be model will occur in phases, minimizing disruption and ensuring a smooth changeover for both staff and clients.

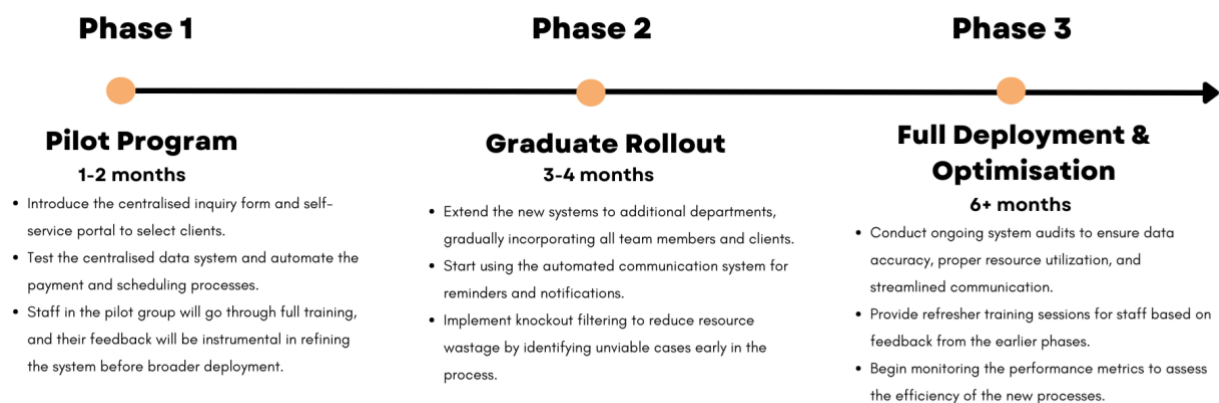


Figure 2. Implementation Phases

Monitoring and Evaluation

Tracking the success of the changes is essential to ensure they meet the intended outcomes. This will involve:

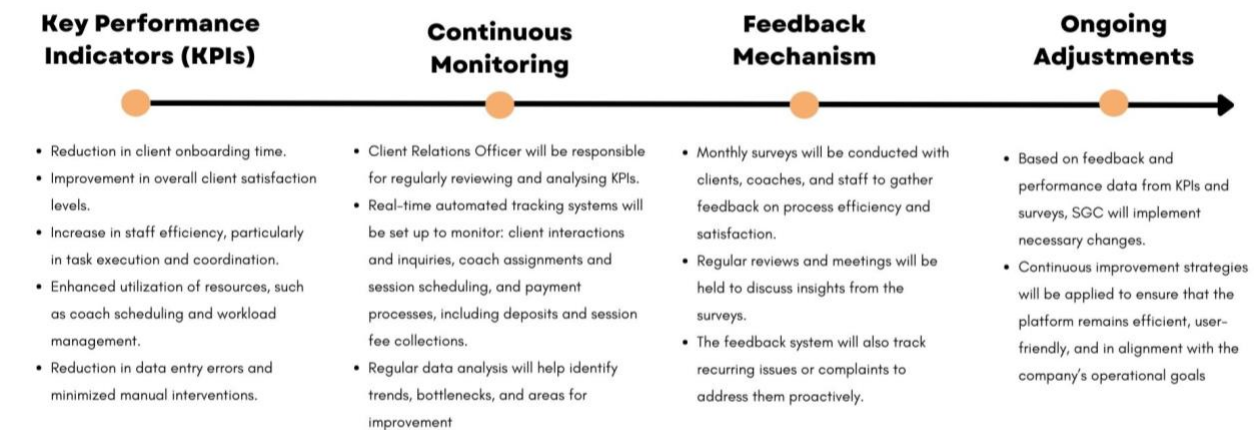


Figure 3. Monitoring and Evaluation

Part 3: Level 2 Business Process Modelling

1) AS-IS Models

The Level 2 AS-IS BPMN diagrams were developed by expanding the Level 1 AS-IS BPMN diagrams from Assignment 1. These diagrams represent the current operational processes of SGC (refer to Appendix D to I for a detailed breakdown and visual representation including explanations).

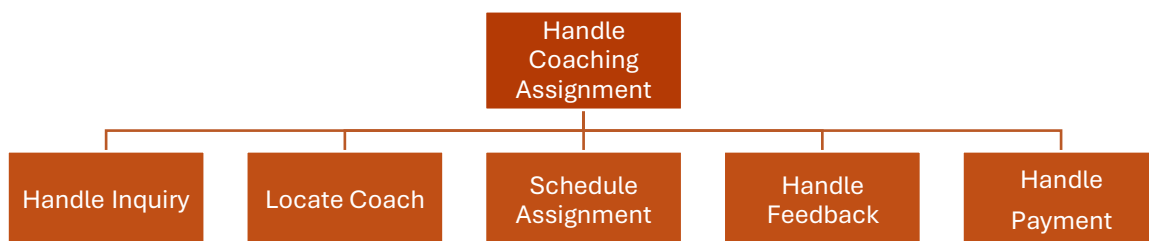


Figure 4. AS-IS Model Structure

2) TO-BE Models

The TO-BE models illustrate SGC's future business processes after implementing proposed improvements. These processes, rooted in the AS-IS models' identified issues, integrate best practices to enhance efficiency. The TO-BE model structure, shown below, promotes a more automated and client-focused approach to process management.

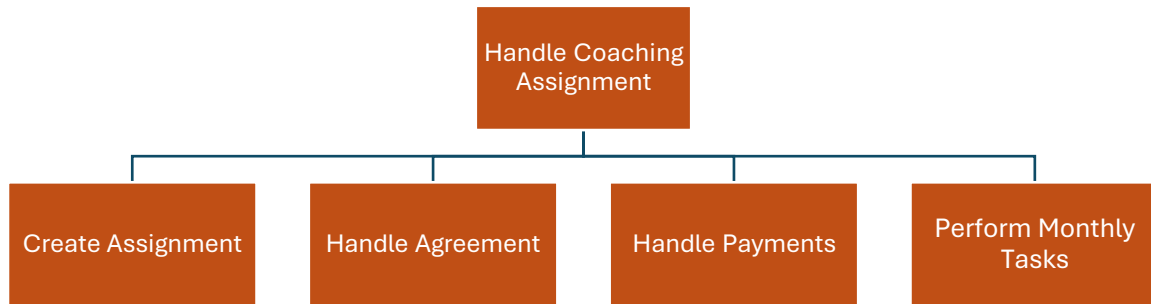


Figure 5. TO-BE Model Structure

For detailed explanations of the TO-BE main process and subprocesses, see Appendices J to O. And an updated overview of the business assumptions, including job descriptions and the updated business model, refer to Appendix C.

Part 4: Automation Model Using BPMS

The current process at SGC places a substantial amount of pressure on the receptionist as they are tasked with a variety of responsibilities, including the manual collection of client information, the drafting of assignments, and the coordination of other team members. This workload frequently results in inefficiencies, delays, and potential human error, which has a detrimental effect on both internal productivity and the client experience (Bourne et al., 2017). Research shows that excessive administrative duties can cause burnout and decreased performance (Nembhard & Uzzi, 2016).

By automating numerous critical tasks, the implementation of an automated business process management system (as shown in Appendix P) would address these concerns (Dumas et al., 2018). This reduces the receptionist's workload, allowing them to focus on higher-value tasks. The automation system improves efficiency, reduces human error, and accelerates client response times (Li et al., 2019), ultimately enhancing operational effectiveness and client satisfaction.

Conclusion

The proposed TO-BE models and automation system for SGC promise significant enhancements, including digitalized client registration, unified data management, optimised resource allocation, and restructured payment processes. These changes aim to minimise manual interventions, prevent bottlenecks, enhance communication, and boost overall efficiency. While these improvements can reduce human error and accelerate response times, critical limitations must be considered. Resistance to new technologies and potential data security risks could hinder implementation. Additionally, automation may decrease personalised client interactions, undermining the personal touch that some clients value. The proposed deposit system might deter new clients unfamiliar with coaching services, potentially placing SGC at a competitive disadvantage as a newcomer without an established reputation. Nevertheless, these improvements lay a strong foundation for enhancing SGC's operations. Maintaining flexibility, continuous monitoring, and a readiness to adjust will be vital for navigating challenges while maximizing efficiency and service quality.

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Appendices

Appendix A. The Devil's Quadrangle

In the dynamic landscape of coaching services, organisations like SGC face the challenge of delivering high-quality solutions while maintaining operational flexibility. As they navigate the complexities of client needs, coach availability, and service quality, understanding their priorities becomes essential for enhancing business processes. SGC places a strong emphasis on quality and flexibility, recognising that client satisfaction is paramount in the coaching industry. While time and cost-efficiency are also important, they are strategically managed to support SGC's overarching goals. By analysing the interplay between quality, flexibility, time, and cost, we can identify key areas for improvement and recommendations that will further strengthen SGC's position in the coaching market.

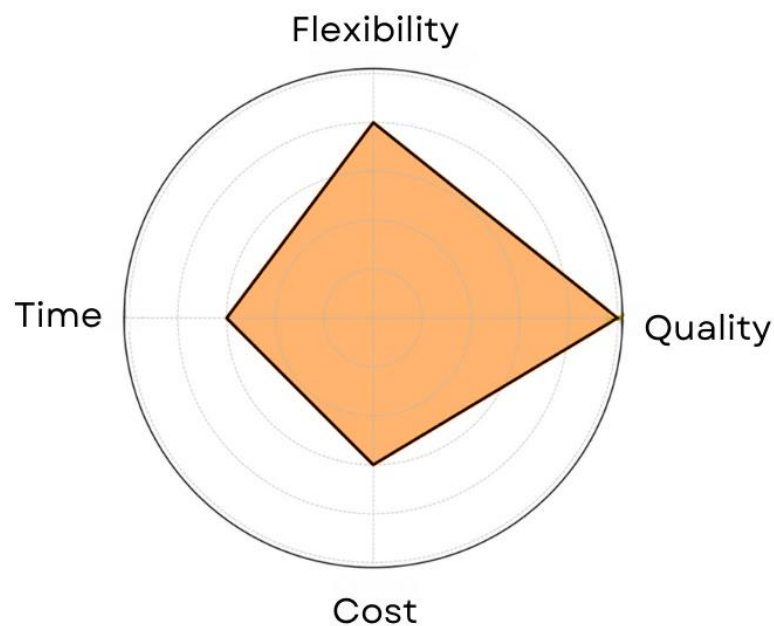


Figure 6. The Devil's Quadrangle

Appendix B. Best Practices Summary

Issue		Best Practices		Application Description and/or Solution Explanation	Advantages	Disadvantages
No	Description	ID	Name			
1	Process Delays Due to Manual Intervention	A	Contact Reduction	Centralized, automated inquiry form on the SGC website that collects client information upfront, reducing manual receptionist tasks.	Reduces delays, improves client onboarding, minimizes manual follow-ups, and enhances client satisfaction	- Requires website updates and may need occasional receptionist support for clients needing help filling the form.
		E	Resource Centralisation	Centralised database for coach information, providing real-time access to profiles, specialties, and availability schedules.	Enhances efficiency, reduces manual updates, improves decision-making, and ensures up-to-date coach details.	Significant upfront investment in technology and training, with potential temporary operational disruptions.
2	Communication and Workflow Inefficiencies	A	Contact Reduction	Automated reminders and notifications, and a self-service portal to consolidate and streamline communication between clients and staff.	Faster response times, clearer exchanges of information, fewer delays, and a more efficient communication process.	Clients may perceive a reduction in direct communication as disengagement, requiring a balance with personalization.
		C	Re-sequencing	Requiring a 20% deposit before a coaching session, with email	Improves payment processing, reduces bad	Upfront deposit may be seen as inconvenient, potentially

				confirmation of session details and payment terms.	debt, and ensures client commitment.	affecting client engagement with SGC's services.
3	Resource Wastage and Ineffective Management	D	Knockout	Proactive case filtering to eliminate non-viable cases early, including rejecting blacklisted clients or unresponsive coaches.	Saves time and improves resource allocation by focusing on viable cases.	Clients rejected early may feel unfairly dismissed, so clear communication is needed regarding knockout criteria.
		E	Resource Centralisation	Unified database for client and coach information, optimizing scheduling and minimizing gaps in coach availability.	Real-time access to information, reduces errors, improves scheduling, and increases operational efficiency.	High initial investment in technology and training, with possible temporary disruptions during transition.
4	Duplication of Efforts and Role Overlap	B	Split Responsibilities	Defined roles and responsibilities within the team, encouraging interdepartmental collaboration to reduce redundancy.	Reduces confusion, improves operational efficiency, and allows staff to focus on priority tasks.	Initial adjustment and training may be required, with some staff resistance to new responsibilities

Appendix C. SGC's New Assumptions

Role/staff	It is assumed that...
Receptionist	• The scheduler in the As-Is model will now become a receptionist, hence 2 receptionists in total. • Receptionists will be mainly responsible for the Create Assignment process • In the beginning of the Create Assignment process, the receptionist will receive a completed form (in which the client has self-completed on the website). If the client calls the receptionist to schedule a session, the receptionist will fill in the form for the client on the website. • During the allocation of coach, if there was a special case of inquiry or coaching demand, the receptionist will contact the client to obtain more information and to clarify with them, via the client's preferred channel of communication. Receptionist then notifies client relation officer about these changes to get updated on future threads of communication. • When the coaching specialist recommends another coach for the client's special case, the receptionist will make suitable changes to the client's assignment on the database. • At the start of every fortnight, receptionist will send out questionnaire for coaches and clients to fill in via their emails. The confirmation email for session and payment already

	includes a link to the survey on the website, which client and coach may wish to fill in anytime.
Coaching specialist	• Coaching specialist will review and recommend new coaches for special cases of coach demand from client. The specialist will be notified by the receptionist. • The coaching specialist will release a form for the coaches to fill in their availability every fortnight, then update the availability of coaches on the website, and make changes to the roster if a coach is unavailable or wish to close their partnership with SGC. • The coaching specialist will seek and hire more coaches every fortnight after reviewing the client database to obtain more suitable coaches to meet the demand of clients.
Client relations officer	• Client relations officer will be mainly responsible for the Handle Agreement subprocess • If the client reaches out and wish to reschedule the session, change the time or cancel the assignment, the officer will manually make changes to the client's assignment on the assignment database, following by notifying a confirmation of change/cancellation to both the clients and the coach.
Bookkeeper	• Bookkeeper will be mainly responsible for the Handle Payment process • The bookkeeper is responsible for compiling invoice, organize coach payment, check, record payment and notify manager if there is a case of debt

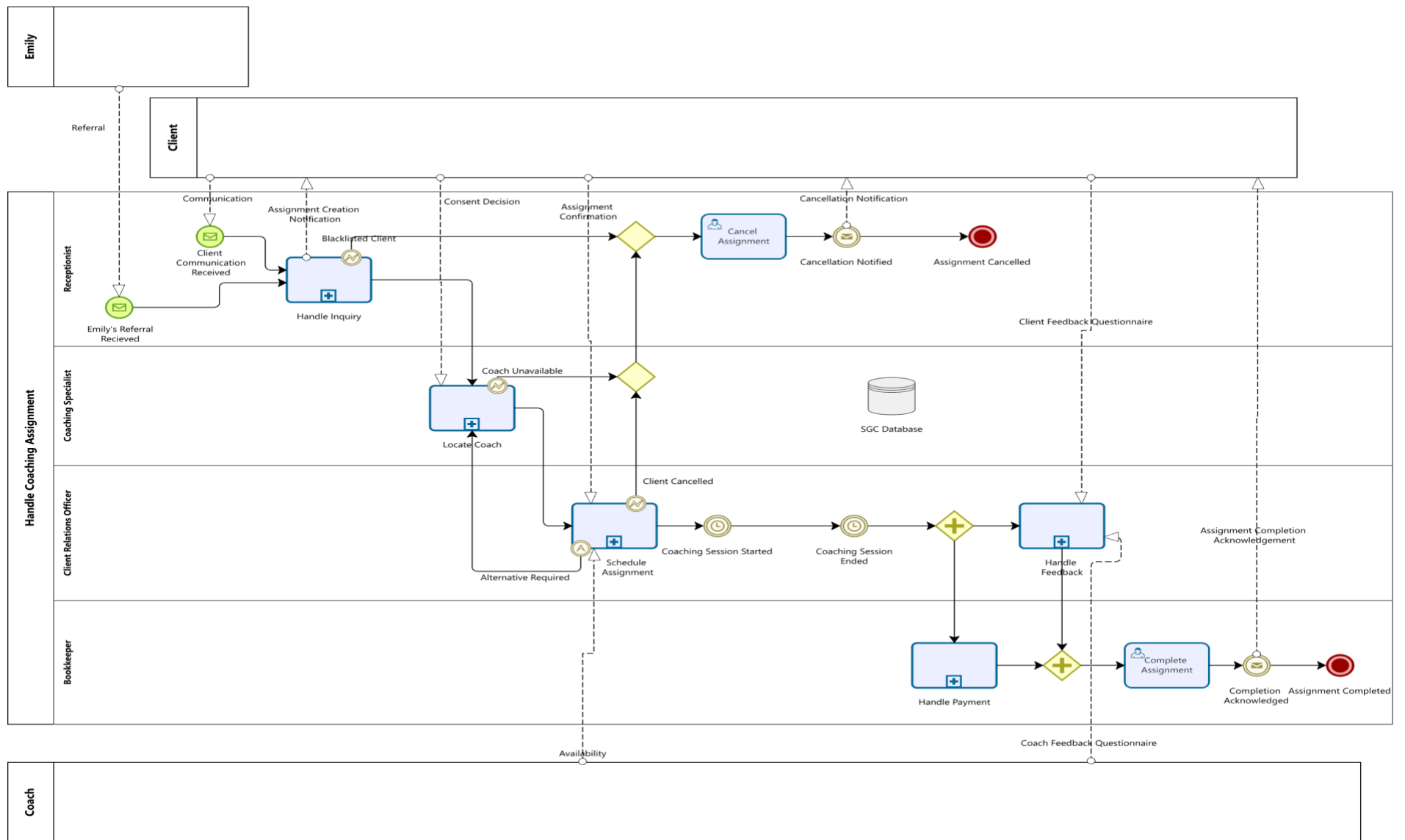
Manager	• Manager is responsible for updating the client database on who is blacklisted and selling debts • Manager is responsible for helping Emily creating the referral code for suitable clients and providing them the codes via email
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Business case	It is assumed that...
Website	• The website has 2 registration form options for new and existing clients. • The website has the coaches' timetable which will be updated fortnightly. • Emily's client has a special referral code to input in the form for discounts.
Registration form	• If the client calls, the receptionist will answer and input client details into the form. • The form has common questionnaires for basic contact information and booking policies.
Email content	• After customer has successfully booked the appointment, an email with agreement policies will be sent, which includes booking status, deposit amount, deposit deadline, terms & conditions, and cancellation policies. • After the appointment, the sent email would include payment invoice and questionnaire.
Questionnaire	Client's questionnaire includes: • Rating overall satisfaction with the session (scale: 1 to 5). • Whether the coach addressed their specific needs and goals (Yes/No).

	• Feedback on the coach’s knowledge and preparedness (Yes/No, with comments). • Rating the professionalism of the coach’s communication (scale: 1 to 5). • Whether the coach listened to concerns and provided helpful feedback (Yes/No, with comments). • Satisfaction with the scheduling process and coach availability (scale: 1 to 5). • Ease of finding a convenient time for the session (Yes/No). • Whether the session helped them move toward their goals (Yes/No, with comments). • Suggestions for improvements in future sessions (open-ended). • Likelihood of recommending the coach and service to others (Yes/No). • Additional feedback for the coach or service (open-ended). Coach’s questionnaire includes: • Rating the client’s engagement and responsiveness during the session (scale: 1 to 5). • Whether the client clearly communicated their goals and expectations (Yes/No). • Whether the coach felt adequately prepared based on the client’s information (Yes/No). • Whether the client’s goals aligned with the coach’s expertise (Yes/No, with comments). • Rating the ease of communication with the client (scale: 1 to 5). • Whether the client’s feedback and questions helped guide the session (Yes/No). • Any issues with scheduling or client availability (Yes/No).
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	• Confirmation that the appointment was conducted as planned (Yes/No, with comments). • Willingness to work with the client again (Yes/No). • Suggestions for improving the process or client experience (open-ended).
Others	• Coach's hourly rate is fixed and predetermined.

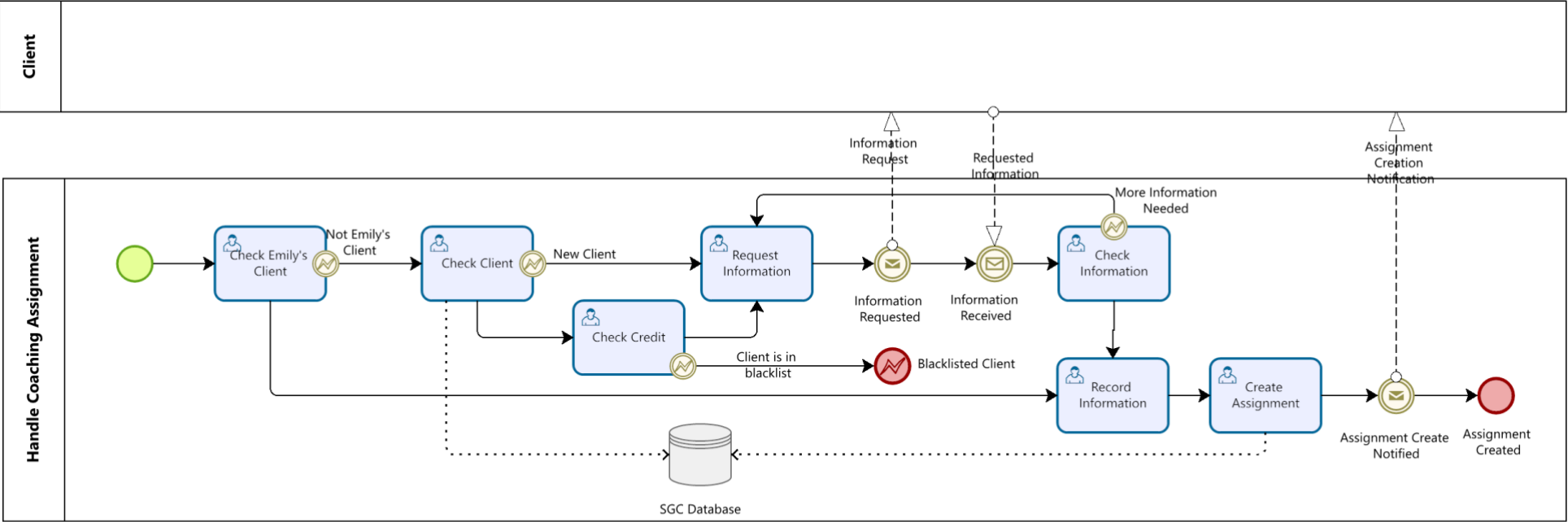
Appendix D. [AS-IS] Handle Coaching Assignment Main Process



[AS-IS] Handle Coaching Assignment Main Process Description

Task/Gateway/Event	Brief Explanation
Emily's referral received (Message Event)	Emily's referral is received, triggering the process to initiate communication with the client.
Client communication received (Message Event)	Confirmation that communication has been established with the client to gather necessary information or verify details.
Handle Inquiry	After verification, the system processes the inquiry to gather the client's personal information and requirements, assess eligibility for Emily's referral discount, and determine if the client is blacklisted or qualified for coaching.
Locate Coach	Identifies an available coach for the client based on the requirements and coach availability.
Cancel Assignment	The assignment is cancelled either due to client choice or unavailability of a suitable coach, and the process terminates.
Cancellation Notified (Message Event)	Notification is sent to all relevant parties that the assignment has been cancelled.
Schedule Assignment	Once a coach is located and confirmed, the coaching session is scheduled based on mutual availability of the client and coach.
Handle Payment	The process to manage the financial transaction for the coaching session, ensuring the payment is handled after the session is completed.
Handle Feedbacks	Collects feedback from both the client and coach, ensuring that the session is evaluated for quality and improvements.
Complete Assignment	Marks the completion of the coaching assignment after feedback and payment processes are finalized.
Completion Acknowledge (Message Event)	Acknowledgment is sent that the assignment has been successfully completed, and the process formally closes.

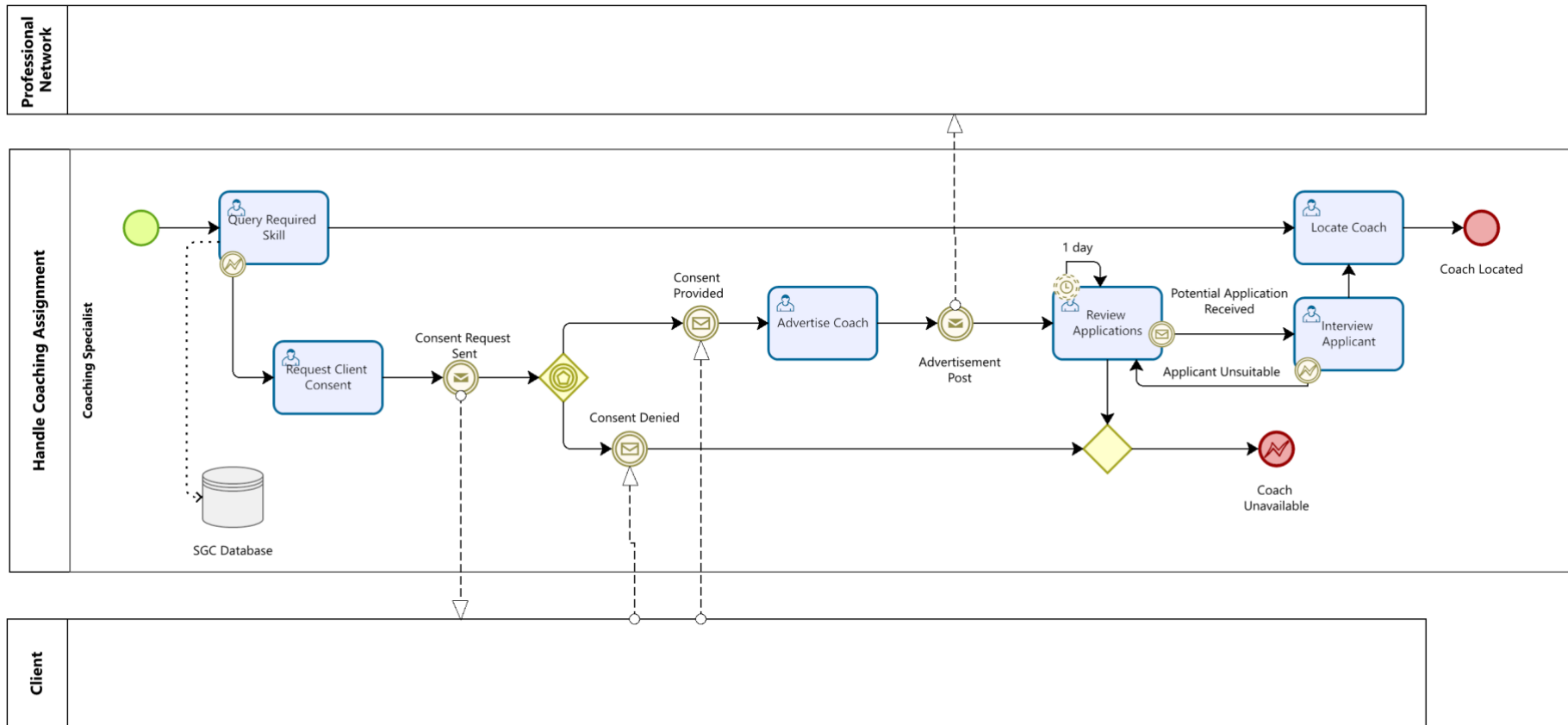
Appendix E. [AS-IS] Handle Inquiry Sub-process



[AS-IS] Handle Inquiry Sub-process Description

Task/Gateway/Event	Brief Explanation
Check Emily's Client	The process starts by checking whether the client requesting coaching is referred by Emily
Check Client	If the client is not referred by Emily, the receptionist will check in the data base to determine if the client is a returning customer
Request Information	If the client is a new client or has good credit, the receptionist asks the client for their personal information, availability, location and requirements for the coaching.
Check Credit	If the client has been served by SGC before, the receptionist will check their credit to determine if the client is in the company's blacklist
(End Event) Blacklisted Client	If the client is in the company's blacklist, the receptionist will cancel the assignment and terminate the sub-process. Otherwise, the receptionist proceeds to request information
(Message Event) Information Requested	The information request is sent to the customer
(Message Event) Information Received	The requested information is provided to the receptionist
Check Information	Once the requested information is received from the client, the receptionist checks if it is sufficient to proceed. If additional clarification is required, the receptionist might request more details, and the process cycles back to request more information.
Record Information	After all the necessary information is received and verified, the receptionist records this data and proceed to create a new assignment
Create Assignment	Once all client information is confirmed and recorded, an assignment is created.
Notify Assignment Creation	A notification is sent to the client, indicating that a new assignment has been created.

Appendix F. [AS-IS] Locate Coach Sub-process

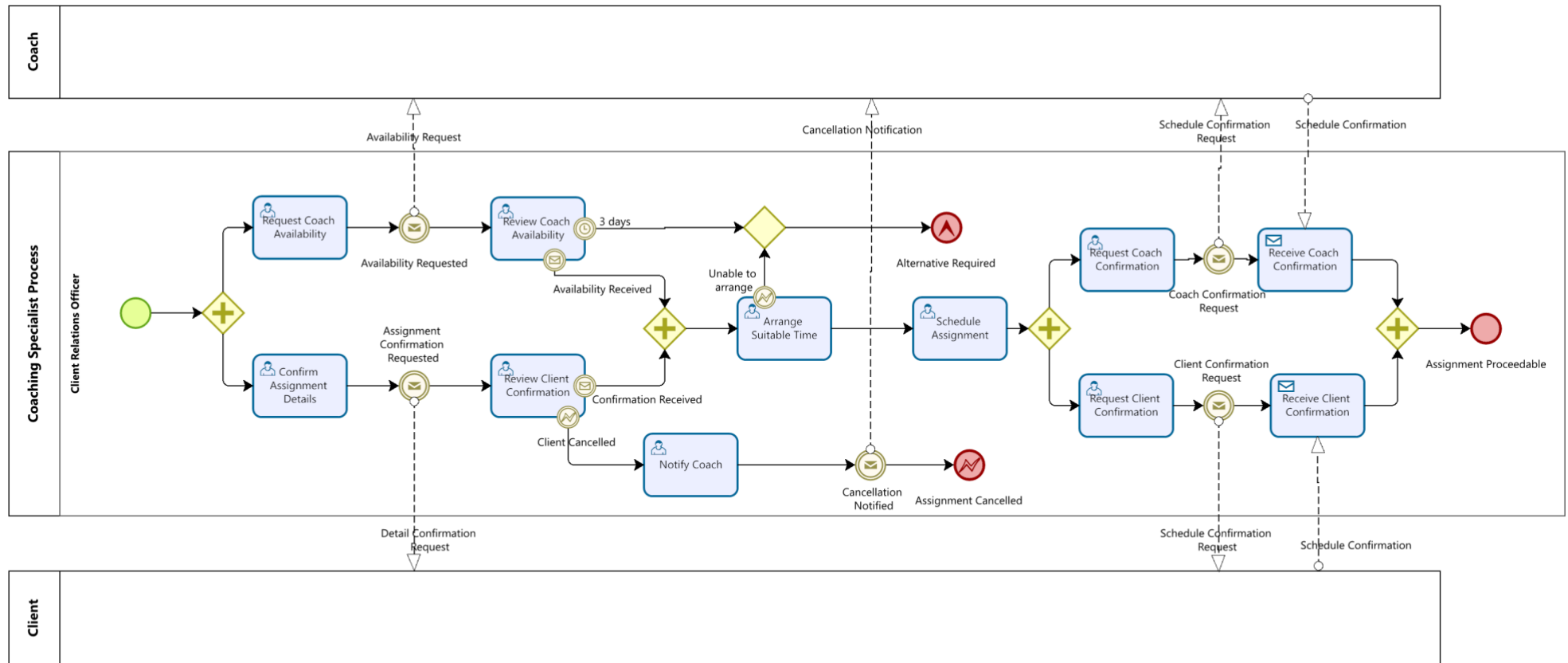


[AS-IS] Locate Coach sub--process Description

Task/Gateway/Event	Brief Explanation
Query Required Skill	The coaching specialist queries the SGC data base to determine the specific coaching skill required by the client
Check Client	If no available coach is suitable, the coaching specialist sends a request for the client's consent to advertise for a coach with their requirements.
(Message Event) Consent Request Sent	A request is sent to the client asking for their consent to begin the advertising process
Consent Provided/Consent Denied	If the client consents, the proceeds to advertise for a coach. If denied, the sub-process ends with coach unavailable
Advertise Coach	With the client's consent, the specialist advertises for a coach with the required skills through professional networks.
(Message Event) Advertisement Post	The advertisement is posted, marking a wait for potential applications from interested coaches.
Review Applications	The coaching specialist reviews incoming applications for the advertised position.
Interview Applicant	The coaching specialist interviews suitable applicants to assess their fit for the client's coaching needs

Record Information	If the applicant is deemed unsuitable, the process returns to reviewing other applications. If no other applications are available, the search for a coach is deemed unsuccessful.
(Attached Event) Application Unsuitable	If the applicant is deemed unsuitable, the process returns to reviewing other applications
Location Coach	Once a suitable coach is found through either the database search or the advertisement, the specialist locates and assigns the coach to the client.
(End Event) Coach Located	The process ends successfully when a coach is located and assigned to the client.
(End Event) Coach Unavailable	If no suitable coach is found after the advertisement and interview process, the process is marked as unsuccessful, and no coach is assigned.

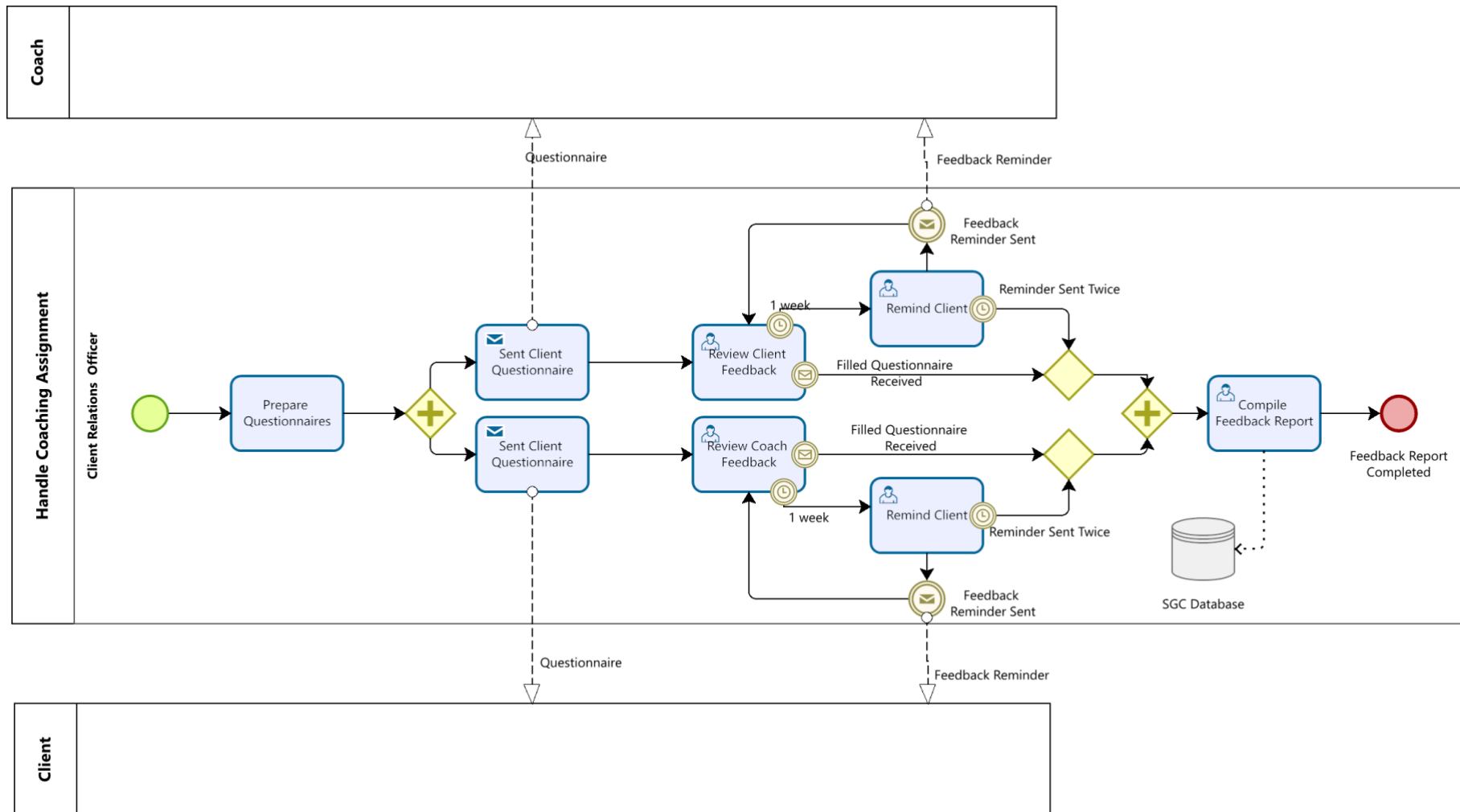
Appendix G. [AS-IS] Schedule Assignment Sub-process



AS-IS] Schedule Assignment Sub-process Description

Task/Gateway/Event	Brief Explanation
Request Coach Availibilty	Initiates a request to check the availability of the coach for the coaching session.
Confirm Assignment Details	Verifies and confirms all necessary assignment details before proceeding with the next steps.
Availability Requested	Sends a formal request to the coach to provide availability details for the coaching session.
Assignment Confirmation Requested	The system sends a request for confirmation of the assignment to both the client and coach.
Review Coach Availability	The coach's availability is reviewed to ensure they can accommodate the coaching session within the requested timeframe.
Review Client Confirmation	The client's confirmation of the session and details is reviewed before moving forward with scheduling.
Notify Coach	If the client cancels the session or is unavailable, the coach is notified of the cancellation.
Arrange Suitable time	Coordinates a mutually convenient time between the client and coach for the session if initial availability does not align.
Schedule Assignment	Finalizes the scheduling of the coaching session after confirming the coach's and client's availability.
Request Coach Confirmation	A request is sent to the coach to confirm their availability and willingness to take on the coaching session.
Request Client Confirmation	A request is sent to the client to confirm the details and schedule of the coaching session.
Coach Confirmation Request (Message Event)	A formal message event where the coach is asked to confirm their participation and availability for the session.
Client Confirmation Request (Message Event)	A formal message event where the client is asked to confirm their attendance and session details.
Receive Coach Confirmation	The system receives confirmation from the coach, agreeing to the session details and timing.
Receive Client Confirmation	The system receives confirmation from the client, agreeing to proceed with the session as scheduled.

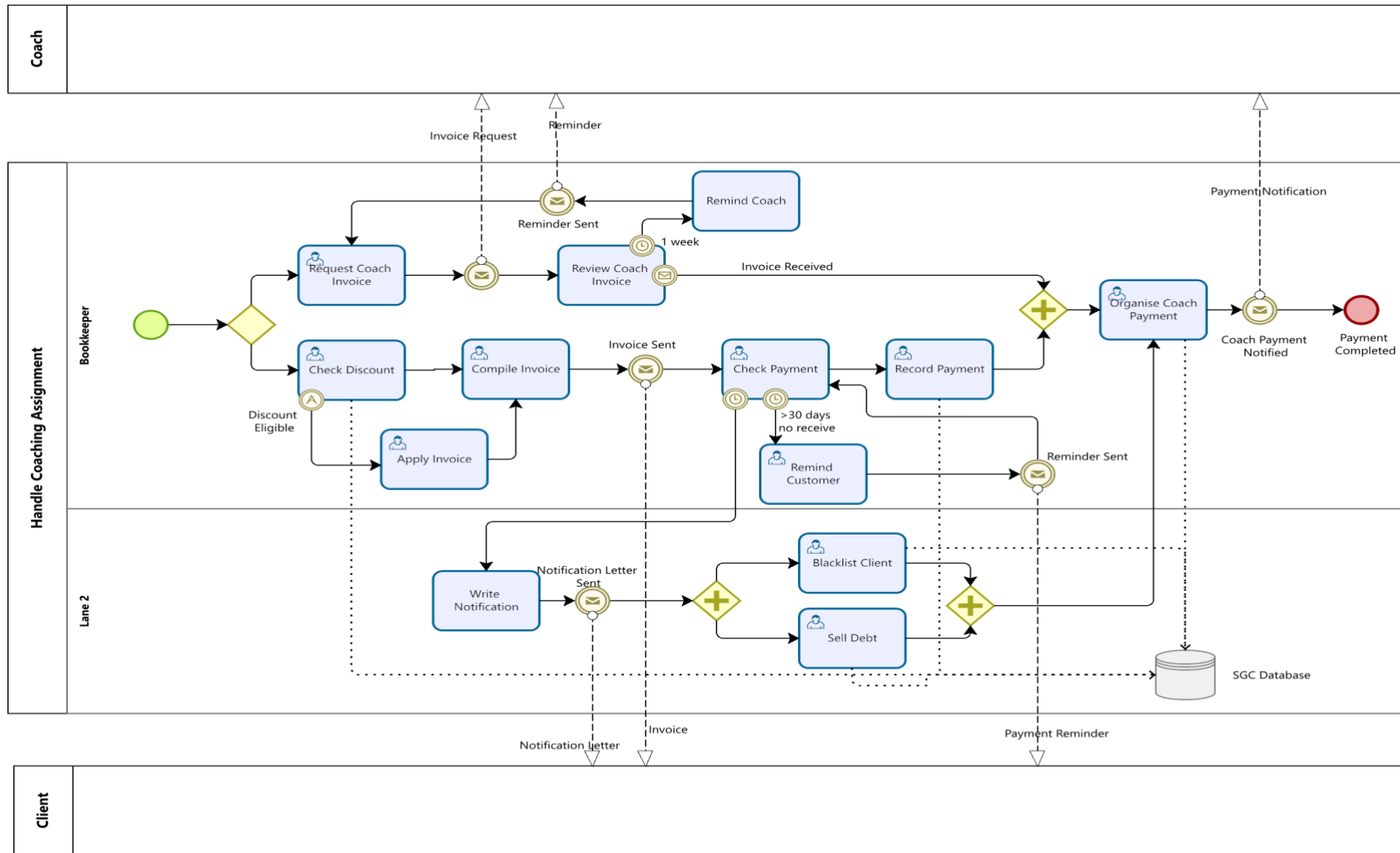
Appendix H. [AS-IS] Handle Feedback Sub-process



[AS-IS] Handle Feedback Sub-process Description

Task/Gateway/Event	Brief Explanation
Prepare Questionnaires	Questionnaires are prepared by the CRO to collect feedback from the client and coach.
Send Client Questionnaire	The questionnaire is sent to the client to gather their feedback.
Review Client Feedback	The CRO reviews client feedback.
Remind Client	A reminder is sent to the client if no feedback is received after 1 week. Another reminder is sent a week after the first reminder if feedback is not received.
Send Coach Questionnaire	The questionnaire is sent to the coach to gather their feedback.
Review Coach Feedback	The CRO reviews coach feedback.
Remind Coach	A reminder is sent to the coach if no feedback is received after 1 week. Another reminder is sent a week after the first reminder if feedback is not received.
Compile Feedback Report	A feedback report is compiled by the CRO based on the responses from the client and coach.
Feedback Report Completed	The final event where the feedback report is completed and stored in the SGC database.

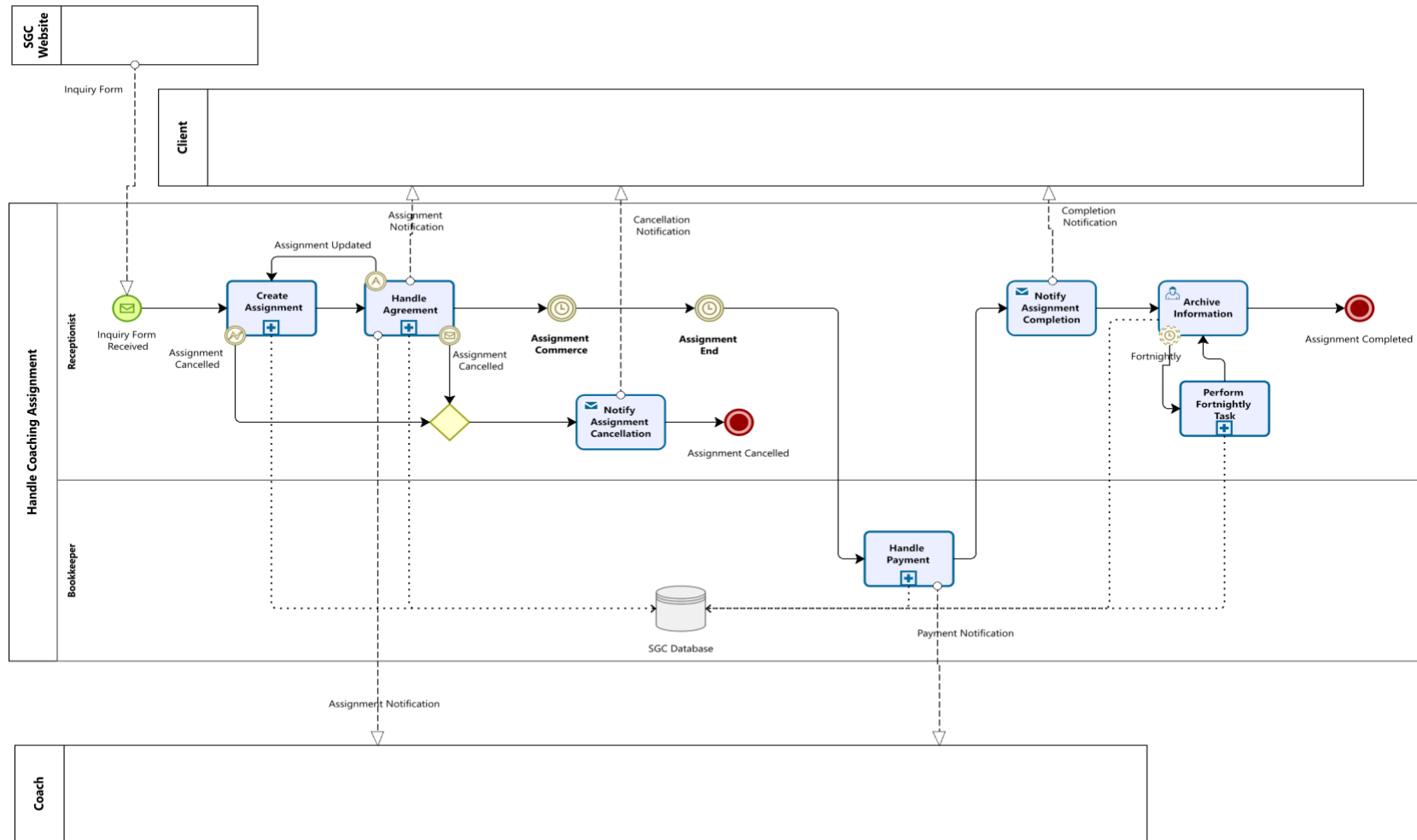
Appendix I. [AS-IS] Handle Payment Sub-process



[AS-IS] Handle Payment Sub-Process Description

Task/Gateway/Event	Brief Explanation
Request Coach Invoice	The bookkeeper requests the coach invoice from the coach.
Review Coach Invoice	The received invoice from the coach is reviewed by the bookkeeper.
Remind Coach	A reminder is sent to the coach if the invoice is not received within a week.
Reminder Sent	
Check Discount	The bookkeepers checks whether the client is eligible for a discount.
Apply Invoice	The invoice is adjusted to apply any applicable discounts.
Compile Invoice	The bookkeeper prepares invoice for the client, including any adjustments.
Invoice Sent	The bookkeeper sends invoice to the client.
Check Payment	The bookkeeper checks whether the client has made the payment after receiving the invoice.
Remind Customer	If payment is not received within 30 days, a reminder is sent to the client. After reminder is sent, the bookkeeper checks until payment is received.
Write Notification	A notification letter is prepared if the payment is overdue for more than 90 days.
Notification Letter Sent	
Blacklist Client	The client is blacklisted if payment is not made after multiple reminders. The outstanding debt is sold to the debt collection agency. Blacklisted client will be recorded into SGC database.
Sell Debt	
Organise Coach Payment	Once the client payment is confirmed / blacklisted, the coach's payment is processed.
Coach Payment Notified	The coach is notified before their payment is processed.
Payment Completed	The process is finalized, marking the payment as completed.

Appendix J. [TO-BE] Handle Coaching Assignment Main Process

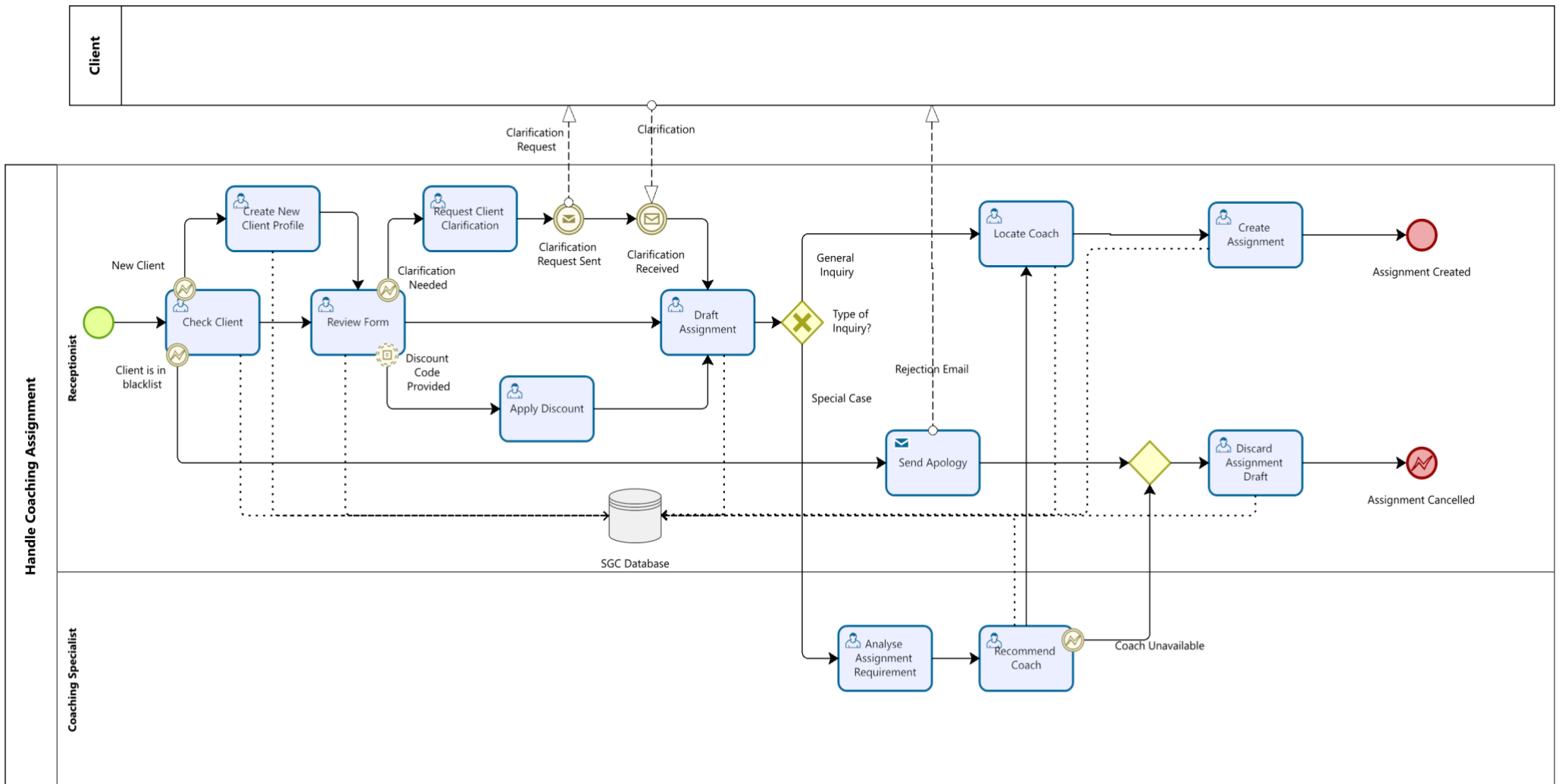


[TO-BE] Handle Coaching Assignment Main Process Description

Task/Gateway/Event	Brief Explanation
Inquiry Form Received (Message Event)	The receptionist receives an inquiry form from the client via the SGC website, initiating the process. Before the process starts, the client must visit the SGC website to fill in the form, which includes personal information, requirements, referral codes, preferred communication platform, and consent signature. Once the client submits the inquiry form, the process officially starts with the "Inquiry Form Received" event.
Create Assignment (Sub-process)	The receptionist creates a new coaching assignment based on the information provided in the inquiry form. This sub-process includes creating the client's profile, reviewing the profile, and applying a referral discount for Emily's clients. In this step, SGC also checks if the client is on the blacklist. If the client is blacklisted, an apology message is sent, and the assignment ends here.
Assignment Cancelled	If any issues arise during the creation or agreement phase, the assignment may be cancelled at this stage.
Handle Agreement (Sub-process)	The receptionist reconfirms assignment details with the client, requests a 20% deposit, and confirms the agreement. If not confirmed, the assignment will be cancelled. If the client requests modifications, the assignment returns to the "Create Assignment" event for updates.
Assignment Commerce (Timer Event)	After the agreement is confirmed, the coaching sessions begin according to the schedule.
Notify Assignment Cancellation	If the assignment is cancelled, the receptionist will send cancellation notification.
Assignment End (Timer Event)	Marks the official end of the coaching assignment once all scheduled sessions are completed.
Notify Assignment Completion	A notification is sent to the relevant parties, informing them that the assignment has been successfully completed.
Handle Payment (Sub-process)	The bookkeeper processes the payment from the client and manages the financial side of the assignment. Payments to the coach are also handled here.
Archive Information	Once the assignment is completed, all related information is archived into the SGC database for record-keeping and future reference.

Perform Fortnightly Task (Sub-process)	Scheduled fortnightly tasks, such as preparing feedback questionnaires for the coach and client, are performed after the assignment ends. This process also includes advertising for new coaches, handling hiring, interviews, and placing the new coach in the SGC centralized database.
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Appendix K. [TO-BE] Create Assignment Sub-process

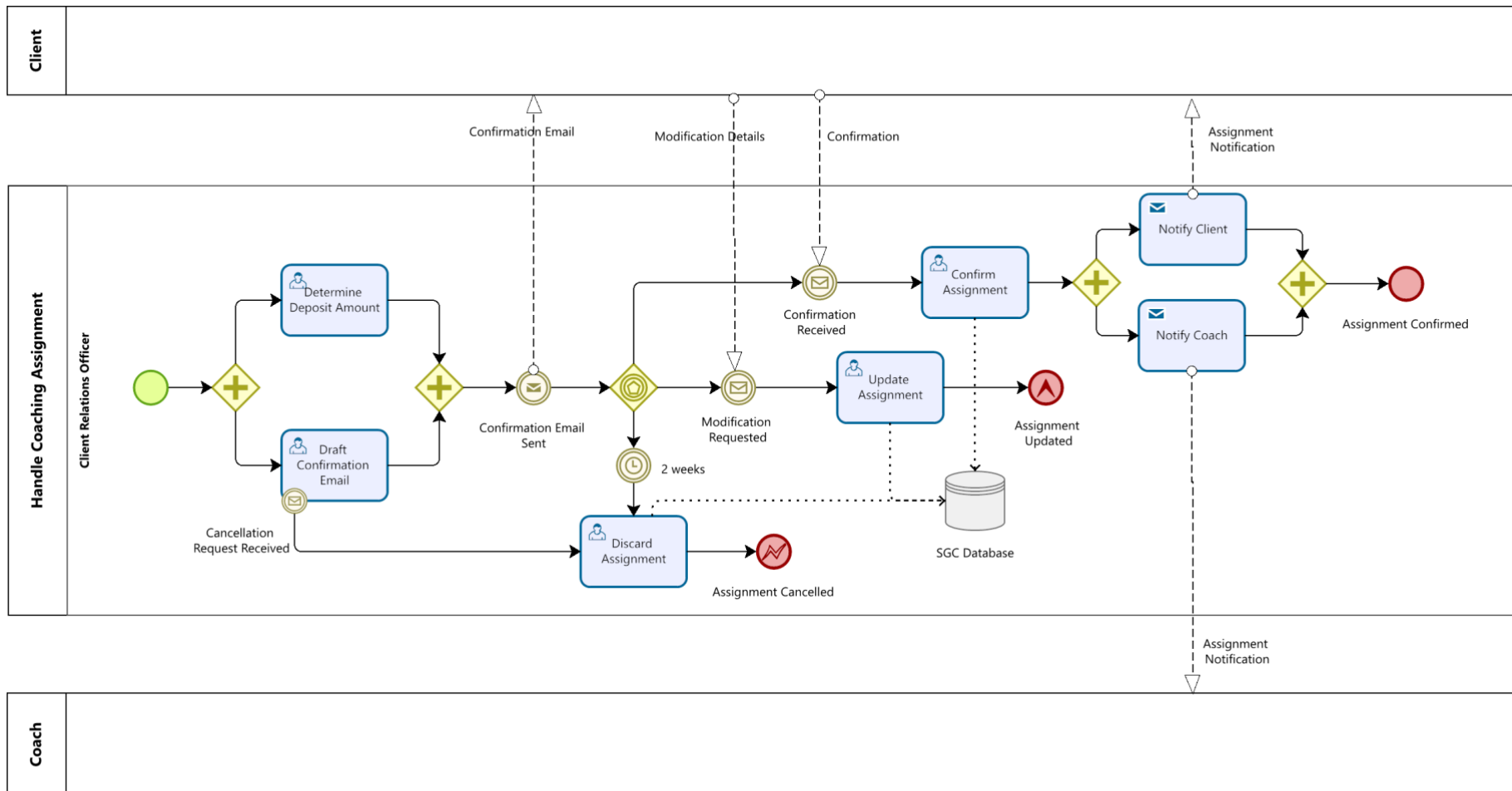


[TO-BE] Create Assignment Sub-process Description

Task/Gateway/Event	Brief Explanation
Start event	Assignment starts when the Receptionist received inquiry of client about service
Check Client	The receptionist checks in SGC database whether the client is new or on the blacklist.
Create New Client Profile	For new clients, a profile is created in the system.
Review Form	The receptionist reviews the form to check if clarification is required or is there any discount available for the customer (through Emily's promo code).
Request Client Clarification	A request is sent to the client for clarification if needed.
Clarification Request Sent	Indicates the receptionist sending a request for clarification to the client.
Clarification Received	Marks the reception of clarification from the client.
Apply Discount	If the form contains an appropriate, it is applied to the client's account.
Draft Assignment	A draft of the coaching assignment is created.
Type of Inquiry?	Receptionist identifies if the is General or Special Inquiry.
Locate Coach	If the inquiry is general, the Receptionist will choose a coach based on his skill based on the information saved in the SGC database.
Analyse assignment	If the inquiry is a special inquiry, it will be sent to the coaching specialist who will review and analyse the inquiry.

Recommend Coach	The coaching specialist recommends a coach for the assignment based on his understanding. If there is no coach available, he notifies the receptionist.
Send Apology	If the client is blacklisted, or no coach is available, the receptionist sends an apology email to the client.
Create Assignment	After a coach is located, the assignment is created and confirmed.
Discard Assignment Draft	If the draft cannot proceed, it is discarded, and the process is cancelled.
Assignment Created	The final event where the assignment is successfully created and confirmed.
Assignment Cancelled	Final event where the assignment is cancelled, stopping the process.

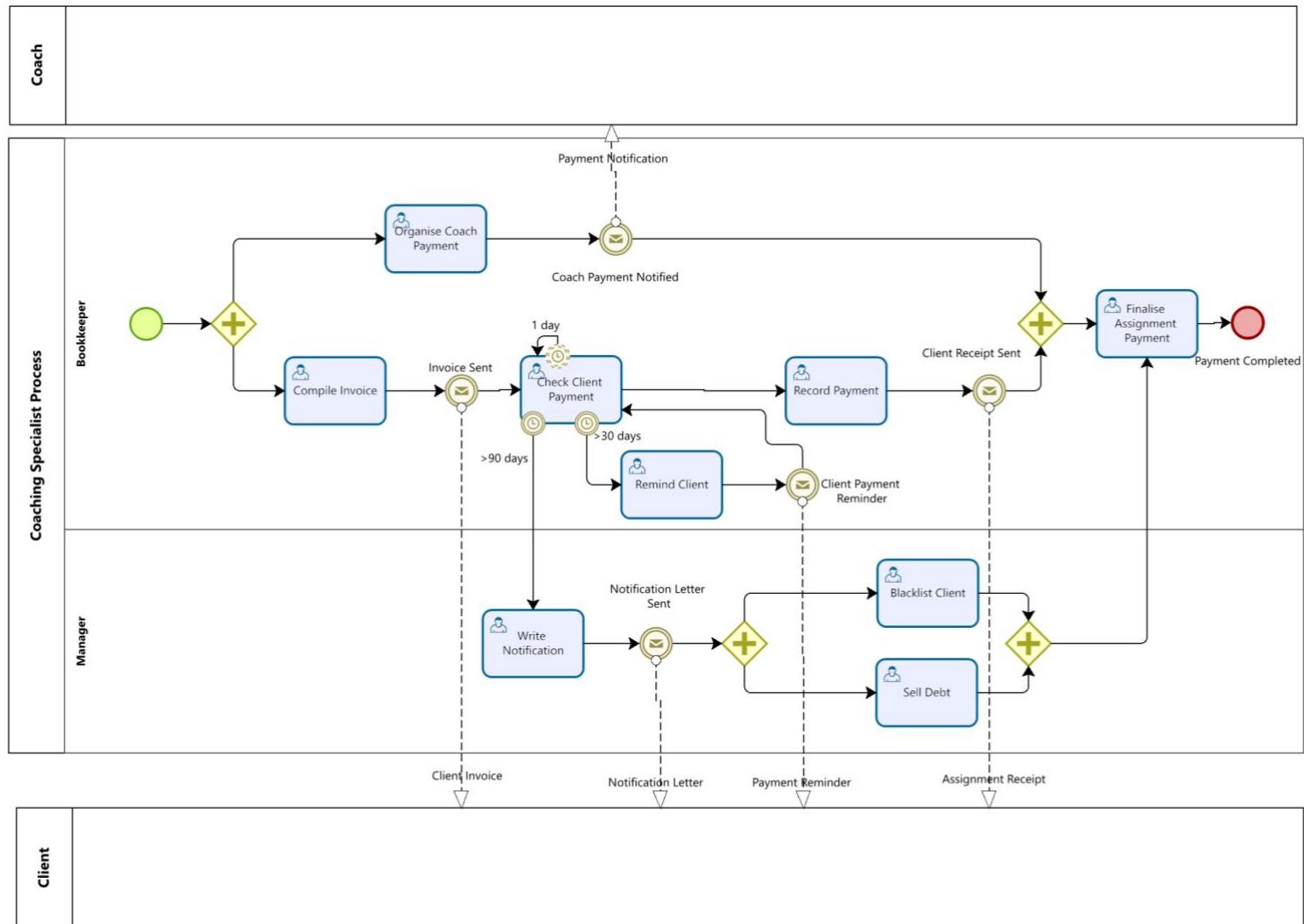
Appendix M. [TO-BE] Handle Agreement Sub-process



[TO-BE] Handle Agreement Sub-Process Description

Task/Gateway/Event	Brief Explanation
Determine Deposit Amount	The client relations officer (CRO) calculates the required deposit for the coaching assignment, and, at the same time, a confirmation email is drafted for the client based on the deposit amount and assignment details.
Draft Confirmation Email	
(Message Event) Confirmation Email Sent	The CRO sends the confirmation email to the client.
(Message Event) Confirmation Received	Confirmation from the client is received regarding the assignment details.
(Message Event) Confirm Assignment	Assignment proceeds and is saved into the SGC database.
(Message Event) Modification Requested	The client requests modifications to the assignment details.
(End Event) Update Assignment	If there is a modification request from the client, the assignment is updated based on the requested modifications by the CRO and saved to the SGC database. The assignment is then escalated
Discard Assignment	If assignment is not confirmed within two weeks or there is a cancellation request from the client, the assignment draft is discarded.
(End Event) Assignment Cancelled	Final event indicating the cancellation of the assignment.
Notify Client	After the assignment details are confirmed, coach and client are informed.
Notify Coach	
(Event Event) Assignment Confirmed	Final event confirming the successful creation of the coaching assignment.

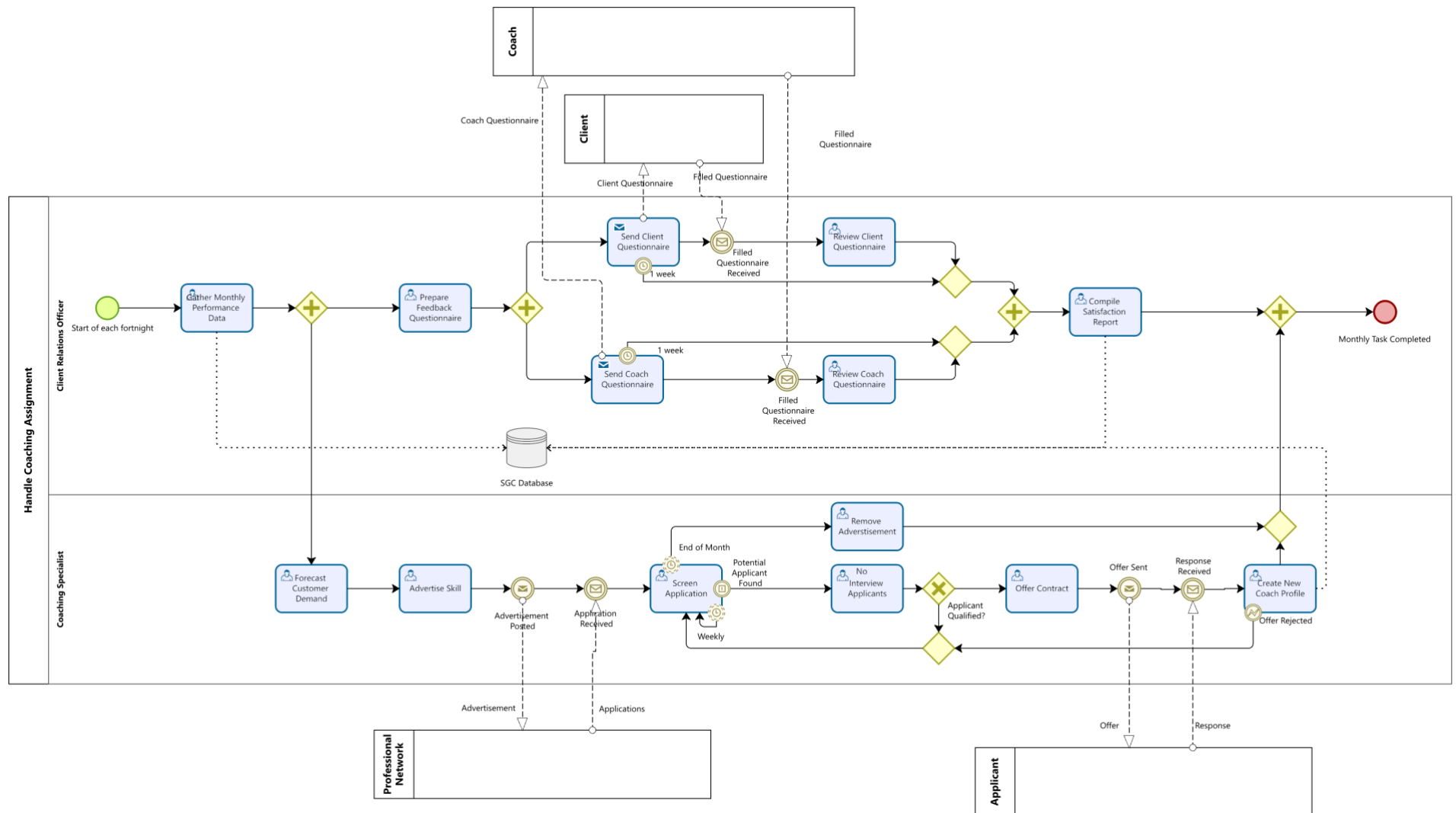
Appendix N. [TO-BE] Handle Payment Sub-process



[TO-BE] Handle Payment Sub-Process Description

Task/Gateway/Event	Brief Explanation
Organise Coach Payment	The bookkeeper organizes payment to the coach.
Coach payment notified	The bookkeeper sends payment details to the coach.
Compile Invoice	The bookkeeper organizes invoice for the client.
Invoice sent	The client received the invoice.
Check Client Payment	The bookkeeper checks for client payment on a daily basis.
Remind Client	A reminder is sent to the client if no payment is received within 30 days.
Client Payment Reminder	Every 30 days, a payment reminder is sent to the client.
Write Notification	A notification letter is written if the payment is overdue by more than 90 days.
Notification Letter Sent	A letter sent to notify the client of overdue payments.
Blacklist Client	The client is blacklisted if payment is not received after multiple reminders, and the outstanding debt is sold to the debt collection agency.
Sell Debt	
Record payment	Payment is recorded into SGC database.
Client Receipt Sent	Assignment receipt is sent to client.
Finalize Assignment Payment	After coach received payment details, and client received receipt, assignment payment is finalized
Payment Completed	Final event indicating that the payment process has been completed successfully.

Appendix L. [TO-BE] Perform Fortnightly Task Sub-process

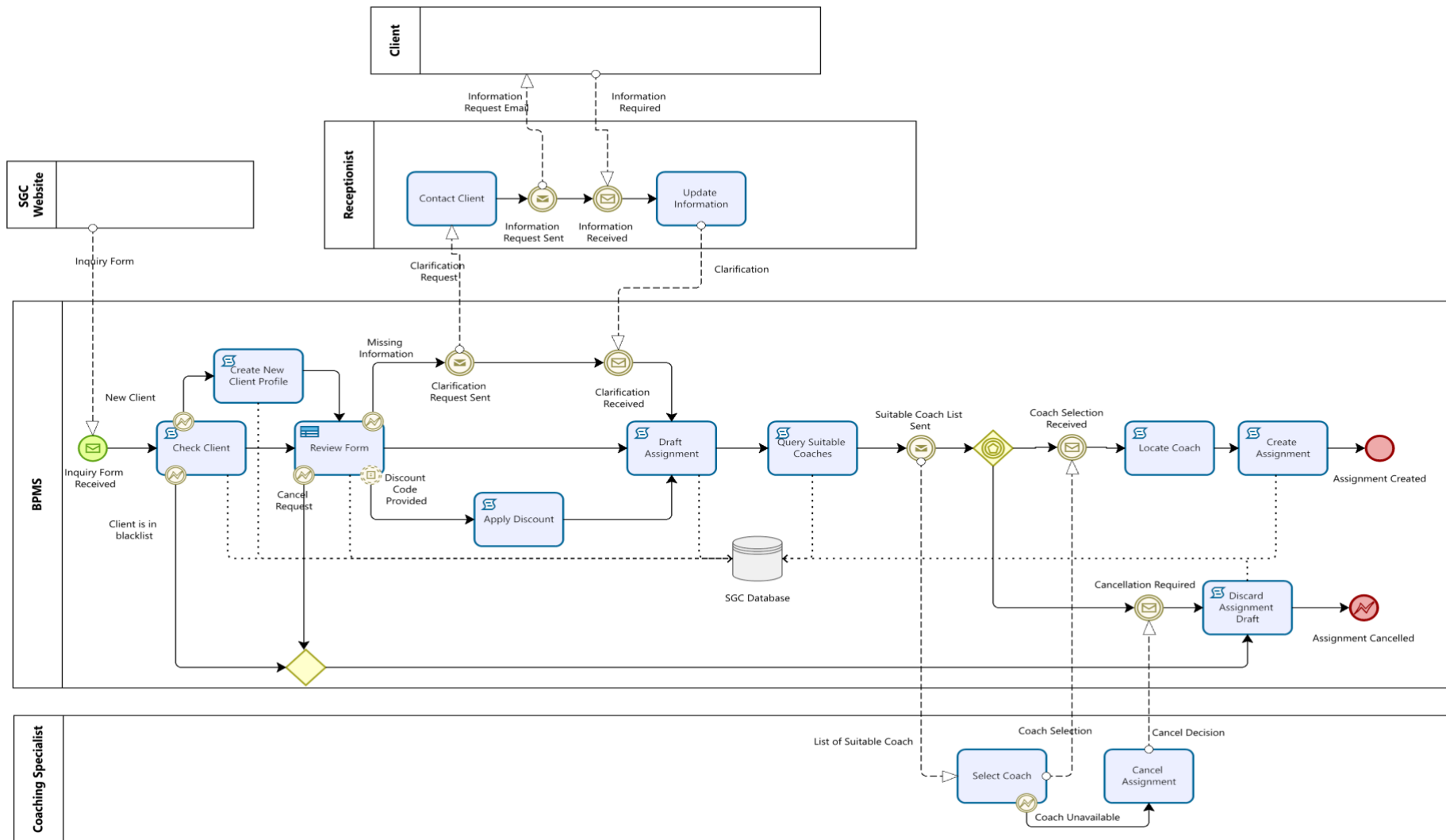


[TO-BE] Perform Fortnightly Task Sub-Process Description

Task/Gateway/Event	Brief Explanation
Enter Monthly Performance Data	At the start of each fortnight, the CRO collect and enter performance data.
Prepare Feedback Questionnaire	The CRO prepare a feedback questionnaire for the client and coach.
Send Client Questionnaire	The feedback questionnaire is sent to the client. After 1 week, regardless the client responded or not, the CRO compiles the satisfaction report
Filled Questionnaire Received	
Review Client Questionnaire	
Send Coach Questionnaire	The feedback questionnaire is sent to the coach. After 1 week, regardless the coach responded or not, the CRO compiles the satisfaction report
Filled Questionnaire Received	
Review Coach Questionnaire	
Complete Satisfaction Report	The CRO compiles a satisfaction report based on the questionnaire reviews and saved into the SGC database.
Forecast Coach Demand	The coaching specialist forecasts coach demand based on past assignment data from the SGC database.
Advertise Skill	The coaching specialist posts job advertisements for coaching applicants with the needed skills.
Advertisement posted	The advertisement is posted on professional networks.
Application received	Coaching specialist receives applications.
Screen Application	The coaching specialist screens applicants on a weekly basis.

Remove Advertisement	The coaching specialist removes the advertisement at the end of every month.
Interview Applicant	If there is a potential applicant, the coaching specialist will interview him. If he fails the interview, the coaching specialist will continue the screening process.
Offer Contract	If the coach passes the interview, he will be offered the contract. The coaching specialist will contact him about contract details, and he will response to the offer. If he rejects, the coaching specialist continues with the screening process.
Offer Sent	
Response Received	
Create New Coach Profile	If the offer is accepted a new coach profile is created and entered the SGC database.

Appendix O. Automation Model



Task/Gateway/Event	Brief Explanation
Start event	The BPMS receives client's inquiry form from the SGC website
Check Client	The BPMS checks client's information from the form to classify if it is new, blacklisted or existing client
Create New Client Profile	If client's information does not match existing rows in current database, a new client row is created
Review Form	Client's information is reviewed to deal with special cases
Clarification Request Sent	If client's information is missing, BPMS sends notification to the Receptionist
Contact Client	After receiving the notification from the BPMS, the Receptionist contact the client for the missing information
Information request sent	The client responded and the receptionist acknowledged the clarification
Information request received	
Update Information	The receptionist enters the missing information to the BPMS
Clarification received	The BPMS logged the complete client information into the database
Apply Discount	If the client's form contains a valid discount code, a discount is applied
Draft Assignment	Assignment is drafted based on client information and stored into SGC database
Query Suitable Coach List	The BPMS filters out coaches who match the client's need to create a list of suitable coaches
Suitable Coach List Sent	Suitable coaches list proceeds to auto-selection / sent to coach specialist to handle special cases. Coach from the suitable list who has the least congested fixture is automatically selected.

Select Coach	Coaching specialist selects most suitable coaches for special cases.
Coach Selection Received	Coach allocation is received
Locate Coach	Coach is allocated
Create assignment	Details of assignment on both client side and coach side is finalised
Assignment Created	Assignment proceeds, process ends
Cancel Assignment	If no suitable coach is available, coach specialist cancels assignment
Cancellation Required	BPMS requires cancellation request to proceed to cancel assignment
Discard Assignment Draft	Assignment draft is discarded as no coaches match the client's requirement
Assignment Cancelled	Assignment is cancelled, process ends

Appendix P. Team Contribution

Tasks	Team Members				
	A: Quy Kim Ngan Tran	B: Duc Nguyen	C: Buu Khanh An Nguyen	D: Trung Kien Le	E: Hong Han Nguyen
Team meetings	100%	100%	100%	100%	100%
Review meetings with teaching staff (preparation, attending, follow up) (Optional)	90%	100%	100%	90%	90%
Team management and administration	100%	100%	100%	100%	100%
Research Searching and Analysing	100%	50%	100%	100%	100%
Construct Process Modelling Analyse the case, development, constructing	50%	100%	50%	50%	50%
Team Documents (e.g. Assignment report) Preparing and Writing	100%	100%	100%	100%	100%
Reviewing Checking and correcting other's work	100%	100%	100%	100%	100%
Percentage of Total Contribution	100%	100%	100%	100%	100%

Member's Name	Members Signature
A: Quy Kim Ngan Tran	Ngan
B: Duc Nguyen	Duc
C: Buu Khanh An Nguyen	An
D: Trung Kien Le	Kien
E: Hong Han Nguyen	Han
End Date	18/10/2024